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PROCESS DESIGN AND CONTROL

Optimal Design and Layout of Industrial Facilities: A Simultaneous Approach

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The design of industrial facilities is often dependent on the layout characteristics. Thus, a simultaneous approach to the design and layout of facilities is an important problem to be addressed. In this paper, the simultaneous design and layout of an industrial facility is studied, taking into account interactions that might be relevant when designing industrial facilities. The model developed here is essentially focused on the layout characteristics, whereas the design aspects are addressed more simply by considering the possible existence of a certain equipment item or connection. This is determined by adequate design models based on the specified industrial design characteristics (as will be explored in part II of this work; see following paper). The proposed model has therefore the particularity of being adjustable to alternative design problems. In this way, different operational and topological problem characteristics can be addressed, often dictated by the type of plant being designed (e.g., flow-shop or job-shop structures and operations such as continuous or batch plants). The optimal plant layout is obtained through the minimization of the connectivity cost, where different topological and operational characteristics are considered, along with equipment costs over a two-dimensional continuous area. The model leads to a mixed integer linear problem (MILP) in which binary variables are introduced to characterize design and topological choices and continuous variables are used to describe the distances and locations involved. To conclude, the applicability of the proposed formulation is illustrated through a set of representative examples.

1. Introduction

The equipment layout decision is an essential stage in the design of industrial facilities. Traditionally, layout issues have been considered a posteriori, once the main plant design stage is completed. However, the interactions of layout with the remainder of the design decisions are often quite strong, which renders a simultaneous approach more desirable.

The layout problem has been extensively studied,^{1–5} although, to the best of the authors knowledge, a simultaneous approach to the design and layout has not been addressed. This lack can be explained by the high level of complexity often associated with each one of these problems separately.

For the treatment of layout independently from global design, the so-called facility layout problem (FLP), four main approaches can be identified in the existing literature. These are the quadratic assignment problem (QAP),⁶ the graph-theoretic approach, heuristics, and finally the mixed integer problem formulation.⁷

In this paper, the simultaneous design and layout problem is addressed by means of a mixed integer approach. The approach presented here is based on a previous work by the same authors developed for the single layout problem.⁵

A generic model is obtained in which the choice of equipment units and connections is handled explicitly when the optimal layout of the plant is determined. No assumptions are made for the equipment items choices (units and/or connections), so as to leave open the possibility of incorporating the optimal layout in different types of design models. The design model is developed using specific structural and operational characteristics of the industrial plants in the study (e.g., multipurpose and multiproduct batch plants or continuous plants, etc.).

Different topological and operational aspects have been considered, such as equipment orientations, distance restrictions, different equipment connectivity inputs and outputs, rectangular and irregular equipment shapes, and space availability; safety and operability restrictions are also contemplated.

The plant is defined within a two-dimensional continuous space, and rectilinear distances are assumed.

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This provides a more realistic estimate of the piping costs as opposed to direct connections.

The final model is a mixed integer linear problem (MILP) in which binary variables characterize design and topological choices whereas continuous variables describe distances and locations.

The optimal plant layout is then obtained for a predefined objective function, which is defined as the minimization of the connectivity cost.

Finally, the applicability of the mathematical formulation is illustrated through the solution of a set of representative examples.

2. Design and Layout Problem Statement

The design and layout of industrial plants as addressed in this paper can be stated formally as follows: Given

- a possible set of equipment items and their geometrical shapes and sizes over a two-dimensional area;
- input and output points locations inside of each equipment space;
- a possible connectivity structure;
- space and equipment allocation limitations;
- safety and operability minimum and maximum distances between equipment items;
- production sections characteristics and associated equipment units involved, if present;
- space availability;
- capital and operational costs of all connectivity structures;
- cost of all equipment units; and
- other design constraints;

Determine

- the final plant equipment choices,
- the optimal plant equipment arrangement (including coordinates and orientations), and
- the associated connectivity structure (inputs and outputs)

so as to optimize a given quantitative objective function while fulfilling all of the constraints defined. In the present case, the minimization of the plant layout equipment and connectivity costs is assumed.

3. Design and Layout Characteristics

The design of industrial plants is, in itself, a complex problem whose options depend on the type of plant under consideration (e.g., flow shop, job shop, etc.). Thus, the choices involved in the design of a multipurpose plant are markedly different from those required for a dedicated plant.

In this paper, the design aspects introduced are restricted to the possibility that a certain equipment item (e.g., unit or connection) is or is not included in the final plant. Thus, no assumption or decision is made on the choice of the type of equipment, which can be taken into account through the use of appropriate models that can easily be incorporated into the present model. This choice is defined through the value of the variables that model the presence of the equipment items (E_g for units and E_c for connections). Conversely, layout characteristics are explored in detail based on the work of Barbosa-Póvoa et al.⁵

A continuous layout space is considered in which a rectilinear approximation is assumed for distances. Multiple equipment connectivity inputs and outputs are

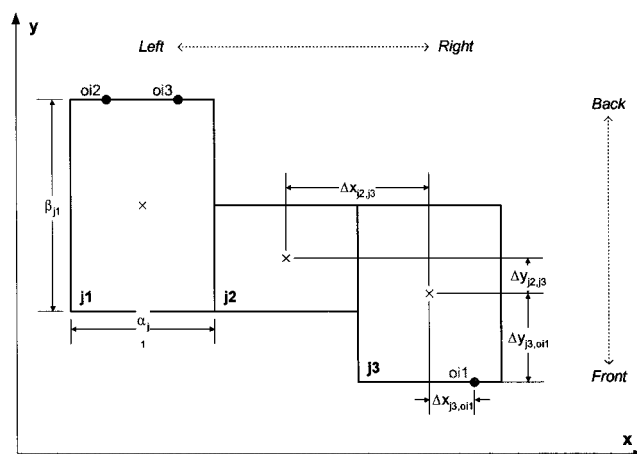


Figure 1. Equipment unit, original position.

considered, as well as space limitations and safety and operability constraints.

Each equipment item is defined as an equipment unit g , and the associated space considers not only the central unit of operation but also all of the associated auxiliary equipment and instrumentation, as well as the space required for maintenance and operation.

Equipment units can have a rectangular or irregular shape. For the latter, a set of rectangular equipment elements j models the entire unit, which has given dimensions over the x and y axes (α_j , β_j) and possible multiple input and output points. An example is shown in Figure 1, where an irregular equipment unit g is considered. It is formed by three rectangular equipment elements $j1$, $j2$, and $j3$, and in its original position, it is defined by a 0° rotation with respect to the x and y axes. Out of three possible combinations, the following two are assumed: $j1$ with $j2$ and $j2$ with $j3$. Each link is established by the definition of the fixed relative distance between the centroids of the two equipment elements. For the pair $j2/j3$ this is expressed as $\Delta x_{j2,j3}$ and $\Delta y_{j2,j3}$, respectively over the x and y axes. The equipment unit depicted has one input point ($oi1$) and two output points ($oi2$ and $oi3$), defined on equipment elements $j3$ and $j1$, respectively. For instance, the input point $oi1$ is fixed by the definition of the relative distance along the x and y axes ($\Delta x_{j3,oi1}$, $\Delta y_{j3,oi1}$) between its location and the location of the centroid of the equipment element $j3$ by which it is circumscribed (see Figure 1).

4. Mathematical Formulation

Using the above design and layout characteristics, the mathematical formulation for the simultaneous design and layout approach is now developed. The indices, parameters, and variables used in this work are defined in the Nomenclature section at the end of this paper.

The following objective function and constraints characterize the generic model for the two-dimensional industrial design and layout problem with variable equipment and connections.

Objective Function. The definition of the objective function (OF) depends on the design model chosen for the simultaneous layout and design of the facility layout problem. The OF adopted was based on the minimization of the total connectivity and equipment capital costs. The former involves the cost of the physical connections and the operating costs caused by material

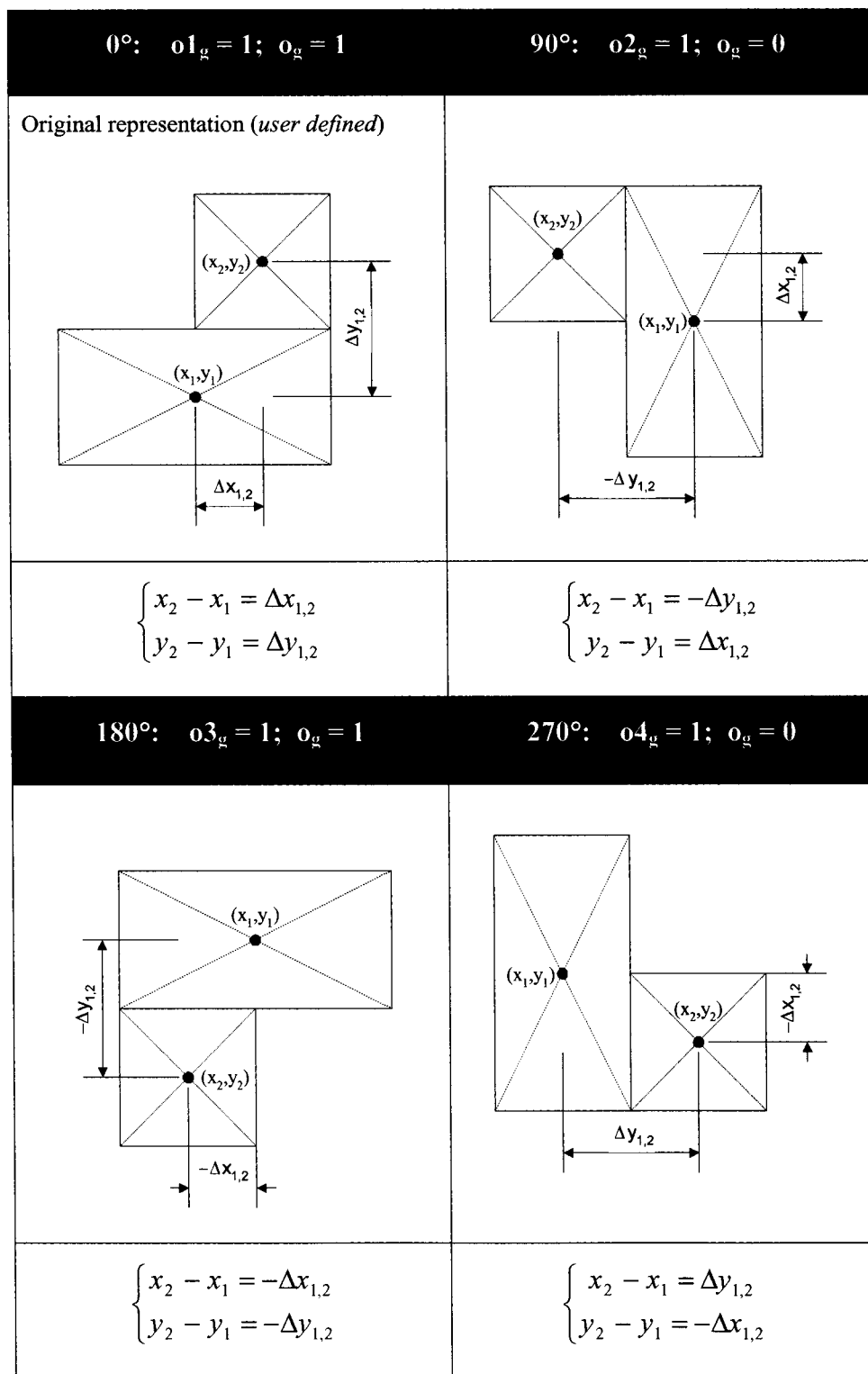


Figure 2. Equipment unit orientation.

transfers occurring within the connection

$$\min \sum_{(oi, oi') | C_{oi, oi'} \neq 0} C_{oi, oi'} D_{oi, oi'} + \sum_g CC_g^0 E_g \quad (1)$$

Constraints. A number of different types of constraints need to be introduced into the model. These include constraints regarding equipment orientation, equipment irregular shapes, input/output locations, connectivity distances, nonoverlapping of equipment, safety and operability, and area allocation.

Equipment Orientation Constraints. For the equipment orientation, it is assumed that rotation is allowed about the x and y axes and is defined through the value of the variable o_g (see Figure 2), if the equipment unit (E_g) is present. According to this definition, the following equations can be written

$$l_j = \alpha_j o_g + \beta_j E_g - \beta_j o_g \quad \forall g, j | j \in J_g \quad (2)$$

$$d_j = (\alpha_j + \beta_j) E_g - l_j \quad \forall g, j | j \in J_g \quad (3)$$

where through the use of the equipment element dimensions (α_j , β_j) and the value of the orientation binary variable (o_g), the equipment element length (l_j) and depth (d_j) are obtained. Two situations can occur depending on the choice of the equipment unit. In the first case, if equipment unit g is present ($E_g = 1$), then for $o_g = 1$, the length of each equipment element j belonging to unit g is equal to α_j (from eq 2), otherwise, for $o_g = 0$, the length is given by β_j . On the other hand, eq 3 states that the depth d_j of equipment element j is equal to the remaining equipment dimension. That is, if $l_j = \alpha_j$, then $d_j = \beta_j$; otherwise, if $o_g = 0$, then $d_j = \alpha_j$. For the second case, if equipment unit g is not chosen ($E_g = 0$), then l_j and d_j both take the value of 0.

Equipment Irregular Shape Constraints. An equipment unit g might have an irregular form dictated from the set of its equipment elements j . Each equipment element j within the equipment unit set J_g is related to the remaining equipment elements through the predefined distances between them ($\Delta x_{j,j'}$, $\Delta y_{j,j'}$). Knowing the predefined set of linked equipment elements pair (Links_g) and assuming that each equipment unit g is at one of the four orthogonal positions allowed ($o1_g$, $o2_g$, $o3_g$, and $o4_g$; see Figure 2), the following equations for two generic equipment units with geometric centers at (x_j, y_j) and $(x_{j'}, y_{j'})$ can be defined

$$x_{j'} = x_j + o1_g \Delta x_{j,j'} - o2_g \Delta y_{j,j'} - o3_g \Delta x_{j,j'} + o4_g \Delta y_{j,j'} \quad \forall g, j, j' | (j, j') \in \text{Links}_g \quad (4)$$

$$y_{j'} = y_j + o1_g \Delta y_{j,j'} + o2_g \Delta x_{j,j'} - o3_g \Delta y_{j,j'} - o4_g \Delta x_{j,j'} \quad \forall g, j, j' | (j, j') \in \text{Links}_g \quad (5)$$

Also, because an equipment unit can only exist in a single position

$$o1_g + o2_g + o3_g + o4_g = E_g \quad \forall g \quad (6)$$

If equipment unit g is present ($E_g = 1$), then one of the above orthogonal binary variables ($o1_g$, $o2_g$, $o3_g$, and $o4_g$) must take a unitary value (binary variable). Therefore, the two equipment elements j and j' are linked by the predefined distances given in eqs 4 and 5. If the equipment unit is not chosen ($E_g = 0$), then its coordinates (x and y) are coincident and not relevant.

The equipment unit orientation is obtained from

$$o_g = o1_g + o3_g \quad \forall g \quad (7)$$

Figure 2 shows the four possible orthogonal positions of an equipment unit with two equipment elements j and j' (1 and 2, respectively). The user must set the original representation (0° rotation) through the definition of parameters $\Delta x_{1,2}$ and $\Delta y_{1,2}$. In the legend are shown the constraints in eqs 4 and 5 applied for each case.

Input/Output Constraints. Different equipment input and output point locations (x_{oi} , y_{oi}) are considered. These points are defined inside each equipment element j . As in the previous constraints, their locations can be easily calculated using the orientation variables, as defined in Figure 3, combined with the coordinates (x_j , y_j) of the geometrical centers of the equipment elements. Thus, along with the constraints in eqs 6 and 7, the following equations are defined

$$x_{oi} = x_j + o1_g \Delta x_{j,oi} - o2_g \Delta y_{j,oi} - o3_g \Delta x_{j,oi} + o4_g \Delta y_{j,oi} \quad \forall g, j \in J_g; oi \in OI_j \quad (8)$$

$$y_{oi} = y_j + o1_g \Delta y_{j,oi} + o2_g \Delta x_{j,oi} - o3_g \Delta y_{j,oi} - o4_g \Delta x_{j,oi} \quad \forall g, j \in J_g; oi \in OI_j \quad (9)$$

where x_{oi} and y_{oi} characterize the coordinates of the different points within the equipment area used for the description of the existence of connectivity. For model simplicity, these points were defined as global in the model, but it is easily understood that they are located within the different equipment elements present (see Figure 3).

Again, if equipment unit g is present ($E_g = 1$), then one of the orthogonal binary variables ($o1_g$, $o2_g$, $o3_g$, and $o4_g$) must take a unitary value. Therefore, the coordinates of the points are determined through the definition of the relative distance ($\Delta x_{j,oi}$, $\Delta y_{j,oi}$) between the points and the geometrical center of the respective equipment element j , that is, eqs 8 and 9. On the other hand, if the unit is not present ($E_g = 0$), its coordinates (x_{oi} with x_j and y_{oi} with y_j) are coincident and not relevant.

Figure 3 exhibits the four possible orthogonal positions of an equipment unit with two equipment elements j and j' , namely, equipment elements 3 and 4 in the example. Point $oi6$ (x_{oi6} , y_{oi6}) is within equipment element 4, and point $oi5$ (x_{oi5} , y_{oi5}) is within equipment element 3. For illustration, the relative constraints (eqs 8 and 9) for the coordinates of point $oi6$ (x_{oi6} , y_{oi6}) are shown in the legend.

Distance Constraints. Rectilinear distances are assumed between the output and input points of each equipment element. In a two-dimensional space, the total distance for each nonzero cost connection (oi , oi') is given by

$$D_{oi,oi'} = Ri_{oi,oi'} + Le_{oi,oi'} + Ba_{oi,oi'} + Fr_{oi,oi'} \quad \forall c, oi, oi' | (oi, oi') \in OI_c \quad (10)$$

where the relative distances (Ri , Le , Ba , and Fr) are obtained from

$$(1 - E_c)M'' + Ri_{oi,oi'} - Le_{oi,oi'} \geq x_{oi} - x_{oi'} \quad \forall c, oi, oi' | (oi, oi') \in OI_c \quad (11)$$

$$Ri_{oi,oi'} - Le_{oi,oi'} \leq x_{oi} - x_{oi'} + (1 - E_c)M'' \quad \forall c, oi, oi' | (oi, oi') \in OI_c \quad (12)$$

$$(1 - E_c)M'' + Ba_{oi,oi'} - Fr_{oi,oi'} \geq y_{oi} - y_{oi'} \quad \forall c, oi, oi' | (oi, oi') \in OI_c \quad (13)$$

$$Ba_{oi,oi'} - Fr_{oi,oi'} \leq y_{oi} - y_{oi'} + (1 - E_c)M'' \quad \forall c, oi, oi' | (oi, oi') \in OI_c \quad (14)$$

$$2E_c \leq E_g + E_{g'} \quad \forall c, g, g' | (g, g') \in G_c \quad (15)$$

where

$$M'' = x^{\max} + y^{\max} \quad (16)$$

Equation 15 implies that a connection c can exist ($E_c = 1$) only if the two associated equipment items (g and g') exist ($E_g = 1$ and $E_{g'} = 1$).

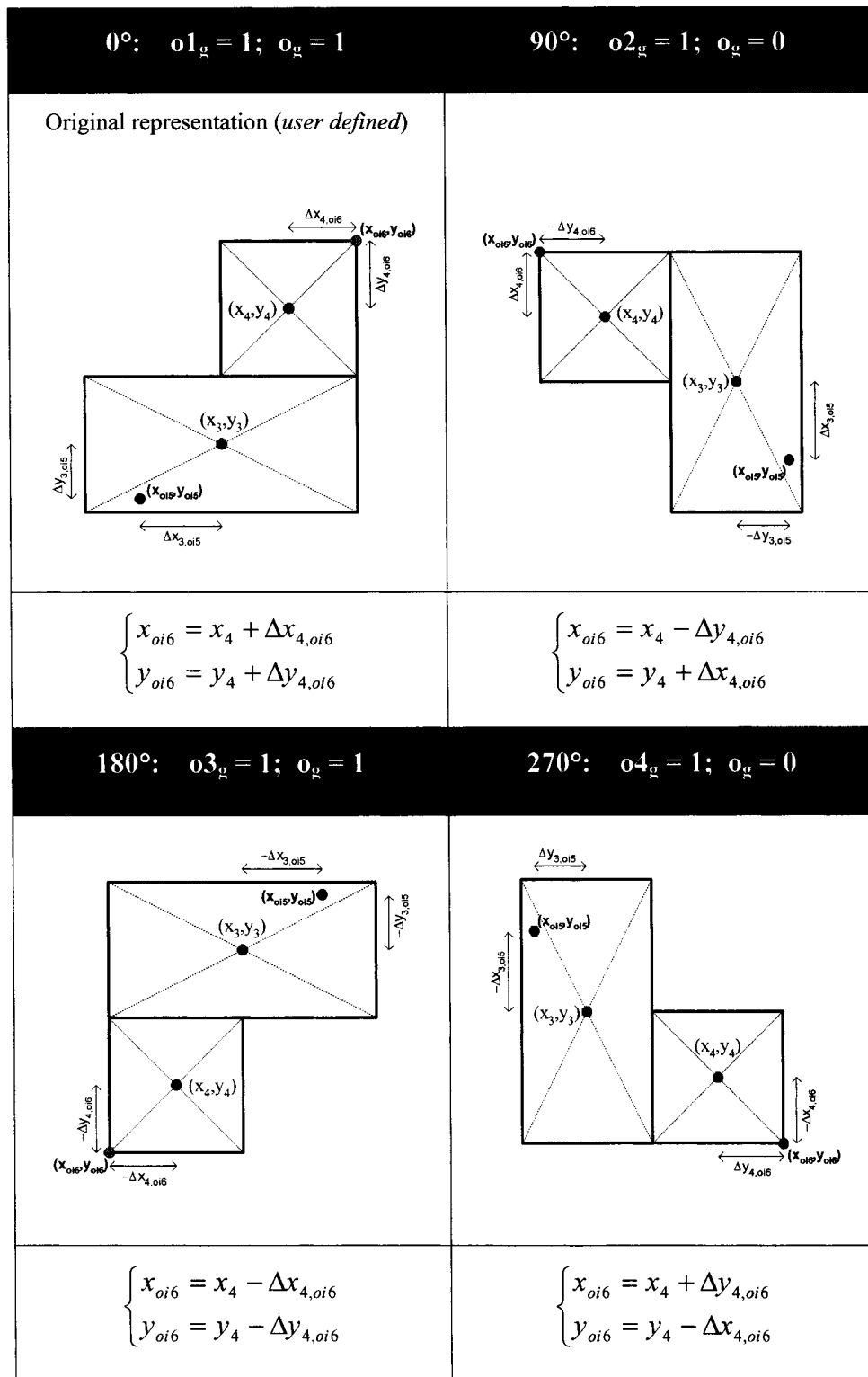


Figure 3. Equipment unit input/output points.

A connection c is defined between the output point oi of an equipment item and the input point oi' of another. If a connection is not present ($E_c = 0$), then eqs 11–14 are inactive because of the very large value of M'' . Otherwise, if a particular connection is installed ($E_c = 1$), then eqs 11 and 12 model the equality $Ri_{oi,oi'} - Le_{oi,oi'} = x_{oi} - x_{oi'}$, whereas eqs 13 and 14 represent a similar equation defined for the y axis, $Ba_{oi,oi'} - Fr_{oi,oi'} = y_{oi} - y_{oi'}$. Therefore, if the output point oi is to the right of

the input point oi' , then x_{oi} is greater than $x_{oi'}$. Thus, eqs 11 and 12 imply that Le (from left) becomes 0 and Ri (from right) takes the positive distance difference between the connection points. The same idea is translated into eqs 13 and 14 in terms of the y axis (Fr denotes front and Ba denotes back).

Nonoverlapping Constraints. Two equipment elements j and j' cannot occupy the same physical location. Thus, a set of disjunctive nonoverlapping constraints

is formulated to guarantee this condition.

$$M^x(2 - E_g - E_{g'}) + x_j - x_{j'} + \frac{M^x(E1_{jj'} + E2_{jj'})}{2} \geq \frac{l_j + l_{j'}}{2} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (17)$$

$$M^x(2 - E_g - E_{g'}) + x_{j'} - x_j + \frac{M^x(1 - E1_{jj'} + E2_{jj'})}{2} \geq \frac{l_j + l_{j'}}{2} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (18)$$

$$M^y(2 - E_g - E_{g'}) + y_j - y_{j'} + \frac{M^y(1 + E1_{jj'} - E2_{jj'})}{2} \geq \frac{d_j + d_{j'}}{2} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (19)$$

$$M^y(2 - E_g - E_{g'}) + y_{j'} - y_j + \frac{M^y(2 - E1_{jj'} - E2_{jj'})}{2} \geq \frac{d_j + d_{j'}}{2} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (20)$$

Note that these constraints are active only if the two equipment elements (j and j') belong to the chosen equipment units (g and g' , respectively). Thus, if either or both of the binary variables that express the units' existence take a null value ($E_g = 0$ and/or $E_{g'} = 0$), the constraints in eqs 17–20 become inactive, because of the large value of the upper bounds M^x and M^y .

Otherwise, if both equipment units are present ($E_g = 1$ and $E_{g'} = 1$), the first term becomes null, and only one of the above constraints can be active (disjunctive condition). This is obtained through the four possible combinations of the values of the two new auxiliary binary variables ($E1_{jj'}$ and $E2_{jj'}$).

The values of M^x and M^y are taken as suitable upper bounds on the distance between any two equipment elements j and j' given by

$$M^x = \min[x^{\max}, \sum_j \max(\alpha_j, \beta_j)] \quad (21)$$

$$M^y = \min[y^{\max}, \sum_j \max(\alpha_j, \beta_j)] \quad (22)$$

Safety/Operability Constraints. Often, minimum and maximum distances between equipment items are defined for safety and operability reasons. To guarantee a certain minimum distance between equipment items, the constraints in eqs 17–20 must be replaced by

$$M^x(2 - E_g - E_{g'}) + x_j - x_{j'} + \frac{(M^x + Zx_{jj'}^{\min})(E1_{jj'} + E2_{jj'})}{2} \geq \frac{l_j + l_{j'}}{2} + Zx_{jj'}^{\min} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (23)$$

$$M^x(2 - E_g - E_{g'}) + x_{j'} - x_j + \frac{(M^x + Zx_{jj'}^{\min})(1 - E1_{jj'} + E2_{jj'})}{2} \geq \frac{l_j + l_{j'}}{2} + Zx_{jj'}^{\min} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (24)$$

$$M^y(2 - E_g - E_{g'}) + y_j - y_{j'} + \frac{(M^y + Zy_{jj'}^{\min})(1 + E1_{jj'} - E2_{jj'})}{2} \geq \frac{d_j + d_{j'}}{2} + Zy_{jj'}^{\min} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (25)$$

$$M^y(2 - E_g - E_{g'}) + y_{j'} - y_j + \frac{(M^y + Zy_{jj'}^{\min})(2 - E1_{jj'} - E2_{jj'})}{2} \geq \frac{d_j + d_{j'}}{2} + Zy_{jj'}^{\min} \quad \forall g, g', j, j' | j' \in J_{g'} > j \in J_g \wedge G_j \neq G_{j'} \quad (26)$$

On the other hand, maximum possible distance restrictions can be modeled as follows:

$$x_j - x_{j'} \leq \frac{l_j + l_{j'}}{2} + Zx_{jj'}^{\max} + M^x(2 - E_g - E_{g'}) \quad \forall g, g', j, j' | (j \in J_g, j' \in J_{g'}) \in \text{Zonx}^{\max} \quad (27)$$

$$x_{j'} - x_j \leq \frac{l_j + l_{j'}}{2} + Zx_{jj'}^{\max} + M^x(2 - E_g - E_{g'}) \quad \forall g, g', j, j' | (j \in J_g, j' \in J_{g'}) \in \text{Zonx}^{\max} \quad (28)$$

$$y_j - y_{j'} \leq \frac{d_j + d_{j'}}{2} + Zy_{jj'}^{\max} + M^y(2 - E_g - E_{g'}) \quad \forall g, g', j, j' | (j \in J_g, j' \in J_{g'}) \in \text{Zony}^{\max} \quad (29)$$

$$y_{j'} - y_j \leq \frac{d_j + d_{j'}}{2} + Zy_{jj'}^{\max} + M^y(2 - E_g - E_{g'}) \quad \forall g, g', j, j' | (j \in J_g, j' \in J_{g'}) \in \text{Zony}^{\max} \quad (30)$$

Again, these constraints (eqs 23–30) are active only if both equipment units are present ($E_g = 1$ and $E_{g'} = 1$).

These constraints can also be defined over specific input and/or output points because of operability conditions such as expedition/dispatch of products where a preferable location for the product storage is desirable so as to guarantee a better material handling (see Barbosa-Póvoa et al.⁵).

Allocation Constraints. Additional constraints must be written whenever the available area is limited to the dimensions of a given industrial facility. Thus, on one hand, lower bounds on the equipment coordinates are defined so as to avoid intersection of equipment elements with the axes

$$x_j \geq \frac{l_j}{2} \quad \forall j \quad (31)$$

$$y_j \geq \frac{d_j}{2} \quad \forall j \quad (32)$$

Similarly, upper-bound constraints are written to force the equipment allocation to lie within a predefined rectangular area confined by $(0, 0)$ and $(x^{\max}, y^{\max})$

$$x_j + \frac{l_j}{2} \leq E_g x^{\max} \quad \forall g, j | j \in J_g \quad (33)$$

$$y_j + \frac{d_j}{2} \leq E_g y^{\max} \quad \forall g, j | j \in J_g \quad (34)$$

If the unit g to which the equipment element j belongs is not selected, both coordinates x_j and y_j , as well as the element's dimensions, take the value of 0.

Table 1. Unit Characteristics

unit	cost (cu)	α	β	input	Δx_{oi}	Δy_{oi}	output	Δx_{oi}	Δy_{oi}
Unit_V1	500	5.0	3.0				OI12	2.5	0
Unit_V2	450	6.0	6.0				OI13	-2	3
							OI14	2	3
Unit_1a	800	6.0	6.0	OI1	-3	0	OI15	3	1
							OI16	3	-1
Unit_1b	900	5.0	5.0	OI2	0	-2.5	OI17	0	2.5
Unit_2a	500	6.0	6.0	OI3	-3	-3	OI18	2.5	2.5
				OI4	3	-3	OI19	-2.5	2.5
Unit_R2	700	4.5	4.5	OI5	0	-2	OI20	0	2
Unit_R4	700	5.0	5.0	OI6	-2	-2	OI21	-2	2
				OI7	0	-2	OI22	2	2
Unit_V5	600	5.0	3.0	OI8	-2.5	0			
Unit_V6	300	6.0	6.0	OI9	0	-3			
Unit_V5a	450	2.0	1.0	OI10	1	0			
Unit_V6a	250	3.0	2.0	OI11	1.5	0			

Table 2. Connections and Associated Costs

connections	oi (output)	oi' (input)	cost (cu/m)
C1	OI12	OI1	1
C2	OI13	OI2	20
C3	OI14	OI5	5
C4	OI16	OI3	10
C5	OI15	OI6	1
C6	OI17	OI4	20
C7	OI20	OI7	5
C8	OI18	OI8	10
C9	OI19	OI9	10
C10	OI21	OI10	1
C11	OI22	OI11	1
C12	OI12	OI2	1
C13	OI12	OI15	1
C14	OI13	OI1	1

Also, space restrictions that lead to a nonrectangular available area can be modeled by defining pseudo-equipment items with fixed sizes and locations that constrain the available area.⁵

In conclusion, the objective function in eq 1 along with an appropriate selection of constraints in eqs 2–34 define the mixed integer linear program (MILP) model for the layout of industrial plants with variable equipment items and connections.

This model can easily be extended to include the presence of possible production/operational sections as shown in Barbosa-Póvoa et al.⁵

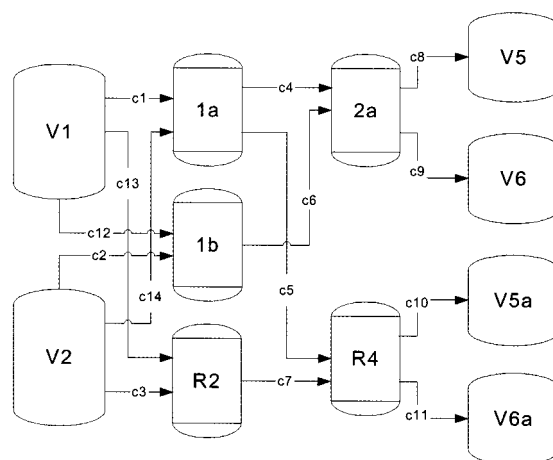
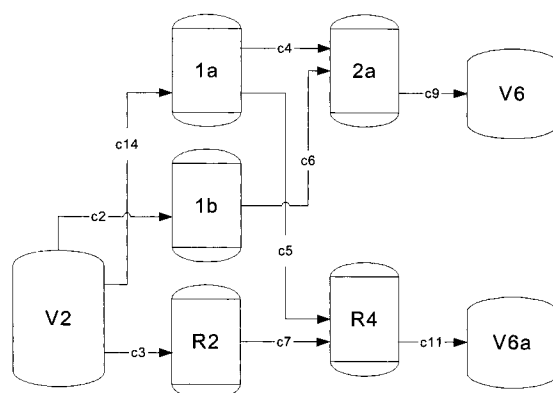
With an adequate generalization, the above simultaneous design and layout model still holds over a three-dimensional space, as presented in Barbosa-Póvoa et al.⁸

5. Example

One of the examples proposed by Barbosa-Póvoa et al.⁵ is studied accounting now for the variable existence of equipment units and connections.

The general algebraic modeling system (GAMS⁹) is used, coupled with the CPLEX optimization package (version 6.5). The problem is solved to the optimum (that is, considering a 0% margin of optimality) on a Pentium II 450-MHz computer taking as the objective function the minimization of the total connectivity costs along with capital costs of the equipment units (eq 1).

Table 1 shows the equipment unit characteristics, and the involved connections and associated costs are presented in Table 2. The plant flowsheet, which is depicted in Figure 4, is to be confined within a predefined rectangular area of $25 \times 25 \text{ m}^2$.

**Figure 4.** Plant flowsheet.**Figure 5.** Optimal plant flowsheet, case 1.**Table 3. Computational Statistics^a**

case	OF	CPU s	nodes	iterations	NIV	NV	NC
1	4634.5	66.31	17 896	124 326	181	349	450
2	4717.5	400.85	81 571	848 154	181	349	450

^a OF = objective function, NIV/NV = number of integer/total variables, NC = number of constraints.

With the purpose of illustrating the model applicability, a set of hypothetical constraints concerning the equipment and connections design were added. In terms of equipment units, it is assumed that only one (disjunctive constraints) of the following equipment pairs could exist: unit V1 or unit V2, unit V5 or unit V6, and unit V5a or unit V6a. On the other hand, all other equipment items must be present. In addition and in relation to the connections, it is assumed that, for a connection to exist, the respective sink and source units must also exist; otherwise, the connection does not exist.

The optimal plant flowsheet is shown in Figure 5, and the corresponding layout is depicted in Figure 6. Vessels V2, V6, and V6a were chosen because they lead to a minimum plant cost (case 1).

The problem statistics and the optimal value of the objective function are reported in Table 3.

To highlight the tradeoff between cost minimization and available allocation area, the same problem was solved for a different plant area, $20 \times 12 \text{ m}^2$, as opposed to the previous $25 \times 25 \text{ m}^2$ (case 2).

The results lead to the optimal plant flowsheet drawn in Figure 7. The associated optimal layout configuration

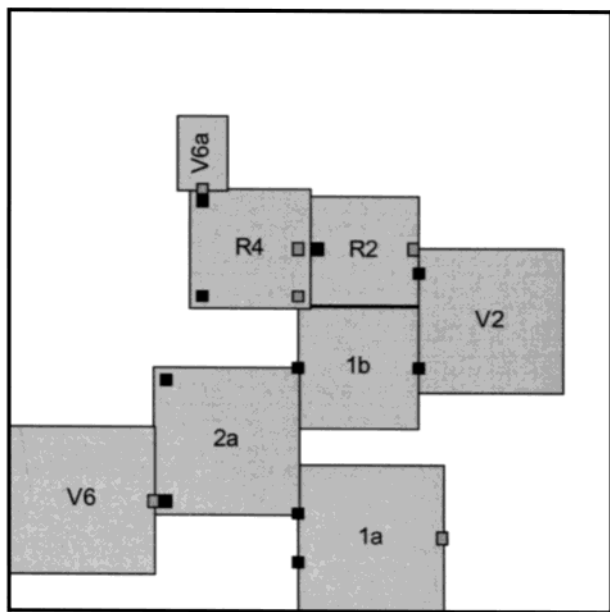


Figure 6. Optimal plant layout (25 m × 25 m), case 1.

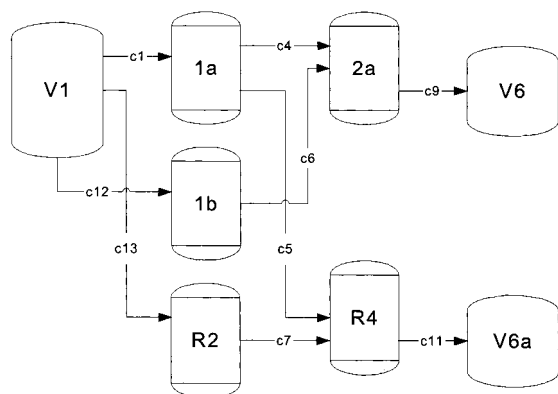


Figure 7. Optimal plant flowsheet, case 2.

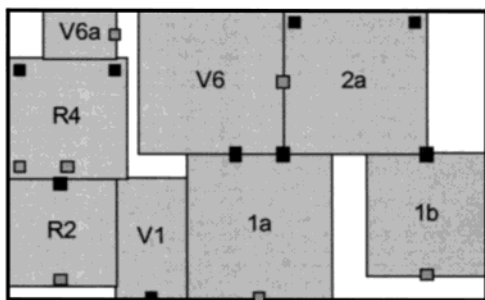


Figure 8. Optimal plant layout (20 m × 12 m), case 2.

is depicted in Figure 8. The computational statistics for case 2, along with the objective function, are given in Table 3.

In comparison with case 1, because of the reduced available allocation area, vessel V2 was replaced by vessel V1 and the value of the objective function increased by 83 cu (50 cu due to the replacement of V2 with V1 and 33 cu from the additional connection costs, where cu denotes currency unit). This shows how the layout restrictions—in this case area availability—can affect the final plant layout and why they should be included, whenever possible, when the plant layout is being determined.

6. Conclusions

A variable layout formulation is presented that is based on a simultaneous approach to the solution of the design and layout problems as a single-level problem. This formulation has the particular quality of being easily adapted to any kind of design problem in which the specific characteristics of a given plant are explored.

The proposed model considers the layout problem over a two-dimensional continuous space and addresses important plant topological aspects, such as different equipment orientations, space availability, different equipment inputs and outputs, and rectangular and irregular shapes, as well as safety and operability restrictions, which are translated into distance restrictions (minimum and maximum).

The model takes the form of a mixed integer formulation that provides the optimal plant design and layout based on a specified economic goal. This goal is defined as the minimization of the costs of connecting structures and equipment units.

In conclusion, it can be stated that a very generic formulation is presented, and the obtained layout results closely describe real-life situations.

Nomenclature

Global Indices

g, g' = equipment unit

j, j' = equipment element

oi, oi' = point (output or input)

c = connection

Sets

$J_g = \{j: \text{set of equipment elements } j \text{ that form unit } g\}$

$G_j = \{g: \text{equipment unit containing equipment element } j\}$
(unique set)

$\text{Links}_g = \{(j, j'): \text{set of linked equipment element pairs } (j, j') \in J_g\}$

$\text{Zonx}^{\max} = \{(j, j'): \text{set of equipment element pairs } (j, j'): G_j \neq G_{j'} \text{ with an upper limit distance between them along the } x \text{ axis}\}$

$\text{Zony}^{\max} = \{(j, j'): \text{set of equipment element pairs } (j, j'): G_j \neq G_{j'} \text{ with an upper limit distance between them along the } y \text{ axis}\}$

$OI_j = \{oi: \text{set of (output or input) points of equipment element } j\}$

$OI_c = \{(oi, oi'): \text{set of output } oi \text{ and input } oi' \text{ pairs defining connection } c\}$

$G_c = \{(g, g'): \text{set of equipment unit pairs } (g, g') \text{ connected by connection } c\}$

Parameters

$C_{oi,oi'}$ = cost per meter of the connection between the output point oi and the input point oi'

CC_g^0 = fixed capital cost of equipment unit g

α_j, β_j = dimensions of equipment element j over the x and y axes, respectively

$x^{\max}, y^{\max}$ = maximum values for the x and y coordinates, respectively

$\Delta x_{j,oi}, \Delta y_{j,oi}$ = relative distance between the point oi and the geometrical center of the equipment element j respectively in the x and y axes, as defined by the original equipment representation (0° rotation)

$Zx_{j,j'}^{\min}, Zy_{j,j'}^{\min}$ = minimum distance allowed between equipment elements j and j' over the x and y axes respectively

$Zx_{j,j'}^{\max}, Zy_{j,j'}^{\max}$ = maximum distance allowed between equipment elements j and j' in the x and y axes respectively

$\Delta x_{jj'}, \Delta y_{jj'}$ = relative distance between the geometrical centers of equipment elements (j, j') : $G_j = G_{j'}$ as defined by its original position, with a 0° rotation over both the x and y axes

Variables

Continuous Variables (All Defined as Positive)

x_j, y_j = coordinates of the geometrical center of equipment element j

x_{oi}, y_{oi} = coordinates of the point oi

l_j = length of equipment element j

d_j = depth of equipment element j

$R_{oi,oi'}$ = relative distance in x coordinates between the output point oi and the input point oi' , if oi is to the right of oi'

$L_{oi,oi'}$ = relative distance in x coordinates between the output point oi and the input point oi' , if oi is to the left of oi'

$B_{oi,oi'}$ = relative distance in y coordinates between the output point oi and the input point oi' , if oi is in the back of oi'

$F_{oi,oi'}$ = relative distance in y coordinates between the output point oi and the input point oi' , if oi is in front of oi'

$D_{oi,oi'}$ = total rectilinear distance between the output point oi and the input point oi'

Binary Variables

o_g = equipment unit g orientation, which equals 1 if the lengths of all equipment elements $j \in J_g$ (parallel to the x axis) are equal to α_j and 0 otherwise

$o1_g, o2_g, o3_g$, and $o4_g$ = equipment unit g anticlockwise rotation, expressed in multiples of 90° ($0^\circ, 90^\circ, 180^\circ$, and 270° , respectively) from the original equipment representation

E_g = equipment unit g existence, which equals 1 if the equipment unit g is present in the solution problem and 0 otherwise

E_c = connection c existence, which equals 1 if the connection c is present in the solution problem and 0 otherwise

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